Year 3 Mathematics

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Learning Strategies

Mathematics is often the most challenging subject for students. Much of the trouble comes from the fact that mathematics is about logical thinking, not memorizing rules or remembering formulas. It requires a different style of thinking than other subjects. The students who seem to be "naturally" good at math just happen to adopt the correct strategies of thinking that math requires – often they don't even realise it. We have isolated several key learning strategies used by successful maths students and have made icons to represent them. These icons are distributed throughout the book in order to remind students to adopt these necessary learning strategies:



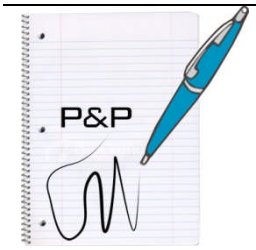
Talk Aloud Many students sit and try to do a problem in complete silence inside their heads. They think that solutions just pop into the heads of 'smart' people. You absolutely must learn to talk aloud and listen to yourself, literally to talk yourself through a problem. Successful students do this without realising. It helps to structure your thoughts while helping your tutor understand the way you think.



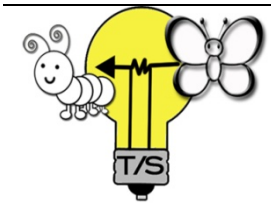
BackChecking This means that you will be doing every step of the question twice, as you work your way through the question to ensure no silly mistakes. For example with this question: $3 \times 2 - 5 \times 7$ you would do "3 times 2 is 5 ... let me check – no 3×2 is 6 ... minus 5 times 7 is minus 35 ... let me check ... minus 5×7 is minus 35. Initially, this may seem time-consuming, but once it is automatic, a great deal of time and marks will be saved.



Avoid Cosmetic Surgery Do not write over old answers since this often results in repeated mistakes or actually erasing the correct answer. When you make mistakes just put one line through the mistake rather than scribbling it out. This helps reduce silly mistakes and makes your work look cleaner and easier to backcheck.



Pen to Paper It is always wise to write things down as you work your way through a problem, in order to keep track of good ideas and to see concepts on paper instead of in your head. This makes it easier to work out the next step in the problem. Harder maths problems cannot be solved in your head alone – put your ideas on paper as soon as you have them – always!



Transfer Skills This strategy is more advanced. It is the skill of making up a simpler question and then transferring those ideas to a more complex question with which you are having difficulty.

For example if you can't remember how to do long addition because you can't recall exactly how to carry the one: $\begin{array}{r} 5889 \\ +\,4587 \\ \hline \end{array}$ then you may want to try adding numbers which you do know how to calculate that also involve carrying the one: $\begin{array}{r} 5 \\ +\,9 \\ \hline \end{array}$

This skill is particularly useful when you can't remember a basic arithmetic or algebraic rule, most of the time you should be able to work it out by creating a simpler version of the question.

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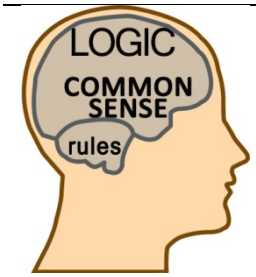
Format Skills These are the skills that keep a question together as an organized whole in terms of your working out on paper. An example of this is using the "=" sign correctly to keep a question lined up properly. In numerical calculations format skills help you to align the numbers correctly.

This skill is important because the correct working out will help you avoid careless mistakes. When your work is jumbled up all over the page it is hard for you to make sense of what belongs with what. Your "silly" mistakes would increase. Format skills also make it a lot easier for you to check over your work and to notice/correct any mistakes.

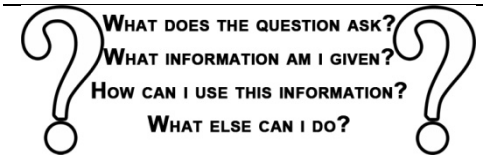
Every topic in math has a way of being written with correct formatting. You will be surprised how much smoother mathematics will be once you learn this skill. Whenever you are unsure you should always ask your tutor or teacher.



Its Ok To Be Wrong Mathematics is in many ways more of a skill than just knowledge. The main skill is problem solving and the only way this can be learned is by thinking hard and making mistakes on the way. As you gain confidence you will naturally worry less about making the mistakes and more about learning from them. Risk trying to solve problems that you are unsure of, this will improve your skill more than anything else. It's ok to be wrong – it is NOT ok to not try.



Avoid Rule Dependency Rules are secondary tools; common sense and logic are primary tools for problem solving and mathematics in general. Ultimately you must understand Why rules work the way they do. Without this you are likely to struggle with tricky problem solving and worded questions. Always rely on your logic and common sense first and on rules second, always ask Why?



Self Questioning This is what strong problem solvers do naturally when they get stuck on a problem or don't know what to do. Ask yourself these questions. They will help to jolt your thinking process; consider just one question at a time and Talk Aloud while putting Pen To Paper.

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Year 3 Mathematics

Number

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Useful formulae and hints

Numbers are written in the form “abc”, where each letter represents a digit

c is the number of ones in the number

b is the number of tens in the number

a is the number of hundreds in the number

For example: the number 325 has 3 hundreds, 2 tens, and 5 ones. These are called the place values of the digits

To group numbers from largest to smallest, work from the left of the number. Compare all the three digit numbers first.

For example: comparing 325, 346, 327, 37, 401, and 53

Of the three digit numbers, there is only one with 4 hundreds; that must be the biggest

If the hundreds digit is the same, compare the tens digits

The next largest number is 346

If numbers have the same hundreds and tens digits, compare their units’ digits.

327 is bigger than 325

Once all the three digit numbers have been compared, do the same for the two digit numbers; 53 is greater than 37

Do the same for single digit numbers if there are any

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To group smallest to largest, follow the above rules but start with the single digit numbers, then two digits, then three

When deciding how to solve word problems, look for key words

More than, together means addition

Less than, difference means subtraction

Times means multiplication

Share means division

When looking for number patterns, work out the difference between two numbers next to each other. See if that rule works for the next two numbers. If it does, use your rule to complete the pattern

Fractions are in the form $\frac{Some\ number}{Some\ number}$

The bottom number is called the denominator and shows the total number of **equal parts** something is broken up into.

The top number is called the numerator, and shows how many of these parts we have

For example, the fraction $\frac{1}{4}$ shows that something is made up of four equal parts, and we have one of these parts

(Think of a cake or pizza)

The chance of something happening can be certain, impossible, or somewhere in between

For example, it is certain that the sun will rise tomorrow, it is impossible that you will turn 200 years old tomorrow, but if you toss a coin you might get heads and you might not

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Exercise 1

Representing Numbers

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1) Write as numbers

a) Twenty seven

b) Forty two

c) Ninety three

d) Twelve

e) Fifty

2) Write as numbers

a) One hundred and three

b) Two hundred and ninety seven

c) Six hundred and thirty three

d) Nine hundred and eleven

e) Three hundred and twenty

3) Write in words

a) 703

b) 297

c) 333

d) 90

e) 201

f) 111

g) 0

4) Write down the number that comes before each of these numbers

a) 33

b) 56

c) 105

d) 12

e) 171

f) 109

g) 243

h) 190

i) 900

j) 30

k) 1

l) 1000

5) Write the number that comes after each of these numbers

a) 19

b) 109

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c) 888

d) 223

e) 801

f) 711

g) 999

h) 309

6) Put these numbers in order from smallest to largest

325, 101, 123, 1000, 946, 121, 15, 221, 323, 104, 694

7) Put these numbers in order from largest to smallest.

201, 204, 402, 912, 911, 333, 322, 921, 221, 121, 4

8) What is the value of the number 4 in each of these numbers?

a) 104

b) 435

c) 214

d) 427

e) 4

f) 40

g) 204

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Exercise 2

Addition & Subtraction

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1) Add these numbers

a) 32 + 14

b) 47 + 19

c) 62 + 35

d) 77 + 22

e) 13 + 17

f) 41 + 44

2) Add these numbers

a) 225 + 52

b) 432 + 41

c) 809 + 77

d) 435 + 23

e) 822 + 11

f) 934 + 73

3) Subtract these numbers

a) 86 - 42

b) 54 - 42

c) 75 -51

d) 99 - 33

e) 54 - 12

f) 65 - 21

4) Peter has 40 cents, John has 25 cents. How much money do they have between them?

5) Alan weighs 45 kg, Chris weighs 48 kg. How much do they weigh together?

6) There are 15 more students in year 3 than in year 4. If there are 46 students in year 3, how many students are in year 4?

7) Tom and Jerry have read 40 books between them. If Tom has read 18 books, how many books has Jerry read?

8) 38 students passed a test, 12 failed, and 5 were absent. How many students are in the class?

9) What number is 43 less than 175?

10) What is the difference between 210 and 344?

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Exercise 3

Multiplication & Division

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1) In each of the pictures below

- How many dots in each row?
- How many rows are there?
- How many dots are there in total?

a)

●●●●

●●●●

●●●●

b)

●●●●●●

●●●●●●

●●●●●●

●●●●●●

●●●●●●

c)

●●

●●

●●

●●

d)

●●●●●

●●●●●

●●●●●

●●●●●

●●●●●

●●●●●

e)

●●●●

●●●●

f)

●●●

●●●

●●●

●●●

2) In question 1, which answers are the same? Why are they the same?

3)

a) How many stars are there in the diagram?

******

******

******

b) How many lots of 6 are there?

c) How many lots of 6 in 18?

d) What is 18 ÷ 6?

e) How many lots of 3 are there?

f) How many lots of 3 in 18?

g) What is 18 ÷ 3?

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4)

a) How many stars are there in the diagram?

****

****

****

****

****

b) How many lots of 4 are there?

c) How many lots of 4 in 20?

d) What is $20 \div 4$?

e) How many lots of 5 are there?

f) How many lots of 5 in 20?

g) What is $20 \div 5$

5) Use the first 4 questions or any other way you know to answer these questions

a) 3×5

b) 5×3

c) $15 \div 5$

d) $15 \div 3$

e) $24 \div 6$

f) $24 \div 4$

g) 7×3

h) 4×8

i) $21 \div 7$

j) $32 \div 8$

6) Multiply the following

a) 9×5

b) 5×9

c) 8×4

d) 4×8

e) 7×6

f) 6×7

g) 3×15 (think of an easier way to do this)

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7) Mary has 4 lollies; Julie has 5 times as many. How many lollies does Julie have?

8) Alan wants to share his lollies amongst himself and his friends so everyone gets the same amount. He has 3 friends and 24 lollies. How many lollies does each person get?

9) Kathy is having a birthday party and her mum wants to make sure there are enough cup cakes for everyone. She thinks each person will eat 3 cup cakes. If there are going to be a total of 11 people at the party how many cup cakes should Karen's mum make??

10) Every child in Tim's class received 4 pencils. If 32 pencils were given out, how many children in Tim's class?

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Exercise 4

Number Patterns

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1) Fill in the blanks

a) 3, 6, 9, ___, 15

b) 2, 4, ___, 8, 10, ___

c) 6, 12, 18, ___, ___

d) ___, 14, 21, 28, 35, ___

e) 4, 8, ___, ___, ___, 24

f) ___, 18, ___, 36, 45

2) Fill in the blanks

a) 25, 20, ___, ___, 5, ___

b) 40, 32, ___, ___, 8

c) 63, 54, 45, ___, ___, 18

d) 63, 60, 57, 54, 51, ___, 45, ___, 39, ___, ___

e) 14, 11, 8, ___, ___

3) Fill in the missing numbers

a) $2 \times 6 = 4 \times __$

b) $5 \times 4 = 2 \times __$

c) $6 \times 6 = 9 \times __$

d) $4 \times 4 = 8 \times __$

e) $7 \times 6 = 6 \times __$

f) $3 \times 13 = 13 \times __$

4) For the given number, list all the numbers that divide into it
Example: 20
1, 2, 4, 5, 10, 20

a) 6

b) 12

c) 16

d) 20

e) 25

f) 7

g) 11

h) What is special about the last two numbers?

5) Jane wants to share her lollies by giving 6 people 5 lollies each. One of the people doesn't want any. How can Jane share her lollies so everyone else gets the same amount?

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6) Tom walked 4 km per day for 6 days in a row. If Alan walks for 8 days, how many km per day should he walk to go the same total distance that Tom did?

7) Peter notices a pattern of fish in a row of fish tanks at the pet store. The first tank had 3 large fish in it. The second tank had 6 medium sized fish. The next tank had 9 smaller fish. There were 7 tanks in the row and the pattern continued to the last one.

a) How many fish were in the last tank?

b) How many fish in the whole row?

8) Graham makes a puzzle for his friends. In a crate he places 84 buttons; in the next one he places 77, then 70 in the next. If he continues this pattern:

a) How many buttons will be in the next crate?

b) How many buttons will be in the last crate?

c) How many crates will he use?

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Exercise 5

Fractions

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1) Write the following as a fraction

a) One half

b) One quarter

c) One eighth

d) Three quarters

e) Five eighths

f) Two quarters

2) Write the following in words

a) $\frac{1}{4}$

b) $\frac{1}{2}$

c) $\frac{3}{4}$

d) $\frac{1}{8}$

e) $\frac{3}{8}$

3) Put these fractions in order from smallest to largest

$\frac{3}{4}, \frac{2}{4}, \frac{4}{4}, \frac{1}{4}$

4) Put these fractions in order from largest to smallest

$\frac{5}{8}, \frac{1}{8}, \frac{7}{8}, \frac{2}{8}, \frac{6}{8}$

5) Fill in the missing numbers

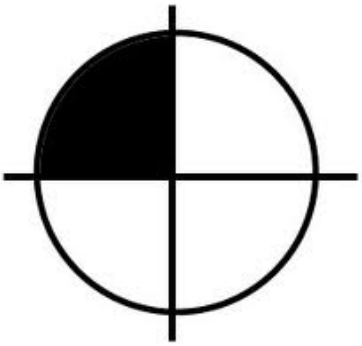
$\frac{1}{2}, \frac{2}{2}, \frac{3}{2}, \frac{4}{2}, __, __$

6) Fill in the missing numbers

$\frac{1}{8}, \frac{2}{8}, __, \frac{4}{8}, __, __$

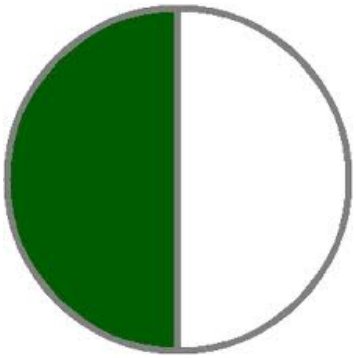
7) What fraction is shaded in the following diagrams?

a)

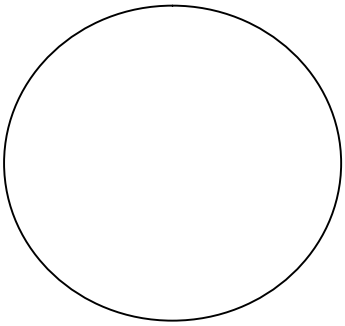


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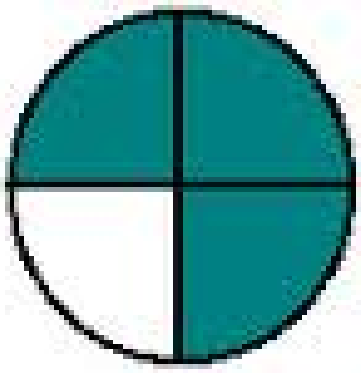
b)



c)

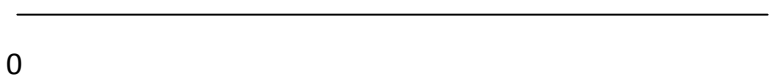


d)



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8) Place the fractions $\frac{1}{2}, \frac{1}{4}\ \frac{3}{4}, \frac{5}{8}, \frac{3}{8}$ on the number line



9) Tim has one quarter of his lollies left, while Jack has eaten three quarters. Who has more lollies left?

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Exercise 6

Chance

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1) Alan tosses a coin. What might the coin show when it lands?

2) Peter rolls a dice. List all the numbers that he could get

3) John has one of every coin in a bag. If he picks one without looking, list what coin he might pull out

4) Veronica has 9 tiles in a bag. Each tile has a different counting number written on it. List what tile she might pull out of the bag

5) There are 6 red shirts, 1 blue shirt and 15 yellow shirts in a draw. If a boy pulls a shirt out without looking:

a) List what colour shirt he might pull out

b) Which colour shirt will he probably pull out?

c) Which colour shirt will he probably NOT pull out?

6) There are 20 red, 5 blue and 1 green lollies in a jar. If Jack closes his eyes and chooses one:

a) What colour lolly will he probably choose?

b) What colour lolly would he be lucky to get?

c) Is he more likely to get a green lolly or a yellow lolly?

d) Name a lolly colour that it would be impossible to get

7) In a jar there are 20 blue buttons. In another jar there are 10 blue and 10 yellow buttons.

a) From which jar would Colin be certain of picking a blue button with his eyes closed?

b) From which jar would be maybe get a yellow button?

c) From which jar would he definitely NOT get a yellow button?

d) Has he got more chance of picking a blue or yellow button from the second jar?

8) Of the following events, which are certain to happen, impossible, or could happen?

a) The sun will rise tomorrow

b) You will eat food

c) You will go to school

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d) You will get every maths question right

e) You will turn 45 years old tomorrow

f) Everyone in your class will win a million dollars tomorrow

g) You will ride a bicycle

9) Tom rolls a normal 6 sided dice. Which number is he most likely to roll?

10) Alan tosses a coin; is it more likely to land on a head or a tail?

11) Peter spins a spinner with 3 red and 3 white faces. Which colour is he more likely to spin?

12) Peter spins a spinner with 1 red and 5 white faces. Which colour is he most likely to spin?

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Year 3 Mathematics

Data

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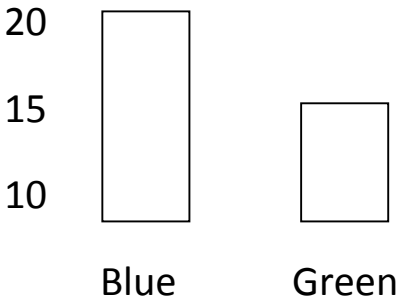
Useful formulae and hints

Data tables show the result of asking or testing something

They show the category (for example favourite colour, favourite car) and the number of people who vote for it or use it

Bar graphs show the same information, but in a form where the higher the bar, the more “votes” the category has received. The number is read on the left of the graph and is equal to the height of the bar. The category is shown under the bar

For example:



Shows that 20 people prefer blue, and 15 prefer green

Picture graphs show similar information but in a way where each picture represents a certain number of “votes”

For example, if a star represented 5 people, then 4 stars would represent 20 people

All three of the above can be used to show the same information. Sometimes it is better to use one type than the other

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Exercise 1

Data Tables

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1) Tom made a table that shows how many of his classmates have each colour as their favourite

Red	Green	Yellow	Blue	White	Black
3	4	1	4	5	2

a) How many children in Tom's class?

b) Which colour was most popular?

c) Which colour was least popular?

d) Which colours had equal numbers of children voting for it?

e) If one child had picked blue instead of white, would that change your answer to part b?

2) A group of people was asked to vote for one day as their favourite day of the week

Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
1	3	5	10	5	6	15

a) How many people were asked?

b) What was most people's favourite day?

c) Why might this be?

d) Which day do most people not like?

3) A man made a list of the cost of a type of blanket at different times of the year

January	March	May	July	September	November
$3.50	$4	$5	$6.50	$5	$4

a) In which of the months was the blanket the cheapest?

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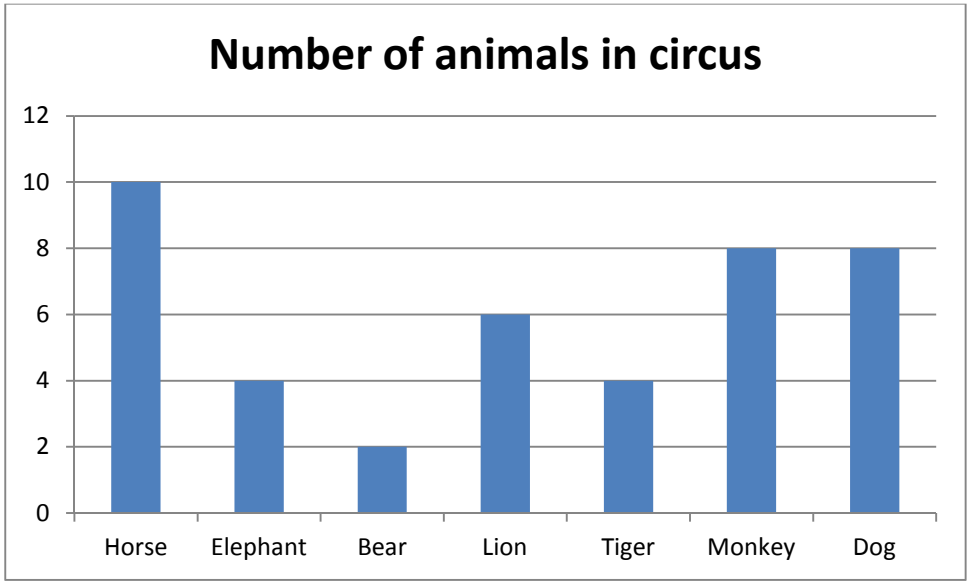
b) In which month was the blanket dearest?

c) What was the difference in its price between these 2 months?

d) Name two months where the price was the same

e) Explain why the price changed so much during the year?

4) The graph below shows the number of different animals in a circus



a) What animal is there most of?

b) What animal is there least of?

c) How many tigers plus bears are there?

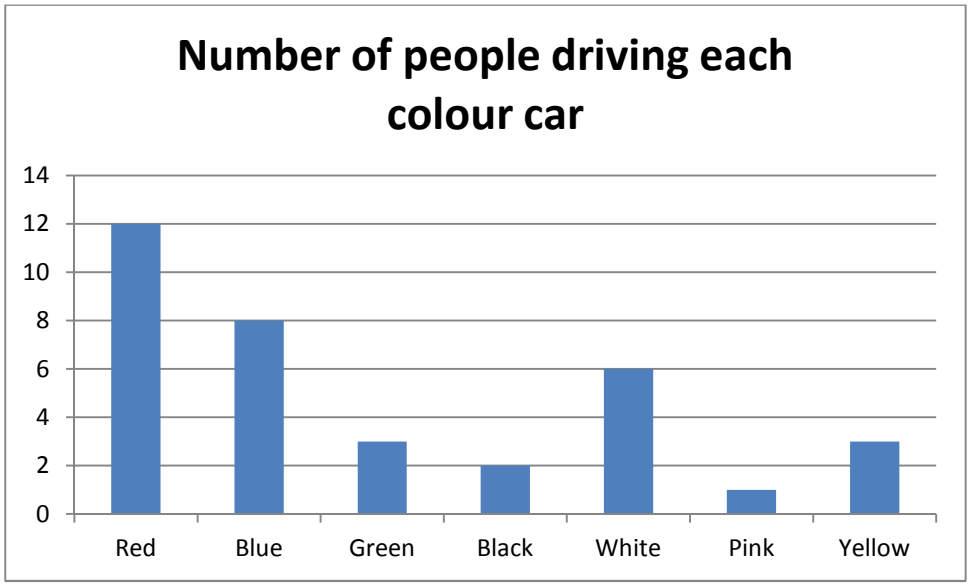
d) How many different types of animals are there?

e) What animal is there exactly 6 of?

f) How many animals in total in the circus?

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5) The graph shows the number of people that own a certain colour car



a) How many people drive a green car?

b) Which colour car do the least number of people drive?

c) How many people were asked the question?

d) Which colour is the second favourite?

e) What colour car do exactly 2 of the people drive?

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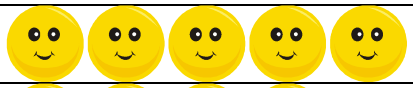
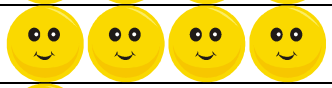
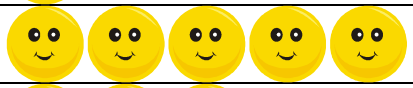
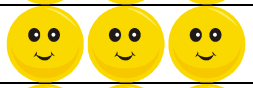
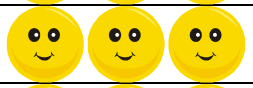
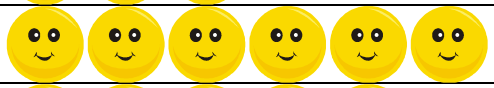
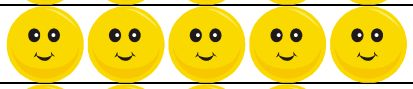
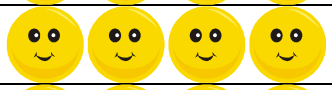
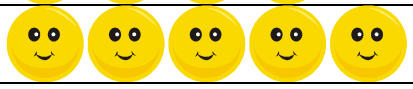
Exercise 2

Picture Graphs

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1) The picture graph below shows a sport and the number of children for whom it is their favourite

Each "face" represents 2 people

Game Number	Attendance
Football	
Rugby	
Soccer	
Basketball	
Hockey	
Swimming	
Tennis	
Golf	
Bowling	
Baseball	

a) Which sport is most popular?

b) For how many people is it their favourite?

c) For how many people is swimming their favourite sport?

d) How many people were asked?

e) Is swimming or hockey more popular?

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2) Some people were asked how many times they ate fish. The picture graph shows their answers. Each fish represents 10 days of the year

Name	Number of days eating fish
Tom	
Benny	
Jane	
Julie	
Karen	
Brian	
Richard	
Ray	
Daniel	
Craig	

a) Who eats fish the most days of the year?

b) How many days a year do they eat fish?

c) Who eats fish on the least number of days?

d) How many days do they eat fish on?

e) If someone ate fish on 45 days of the year, how could you show this on the graph? Can you think of a better way to show numbers of days that are not groups of 10?

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3) A student went to all the shops in his area and found which of the 5 fruits below was most expensive in that shop. Each piece of fruit is a shop where that fruit was most expensive



a) In how many shops were strawberries the most expensive?

b) What fruit was the most expensive in only 2 shops?

c) How many shops did the student visit?

d) What fruit was most expensive in more shops?

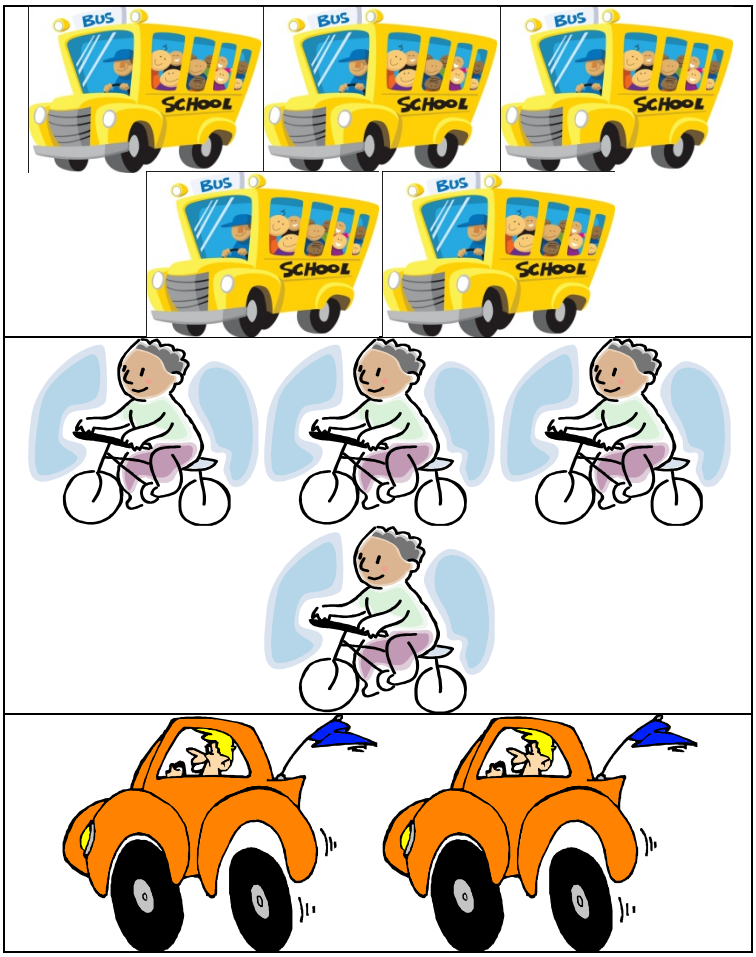
e) What could the student have done if he found a new shop in which pears were the most expensive?

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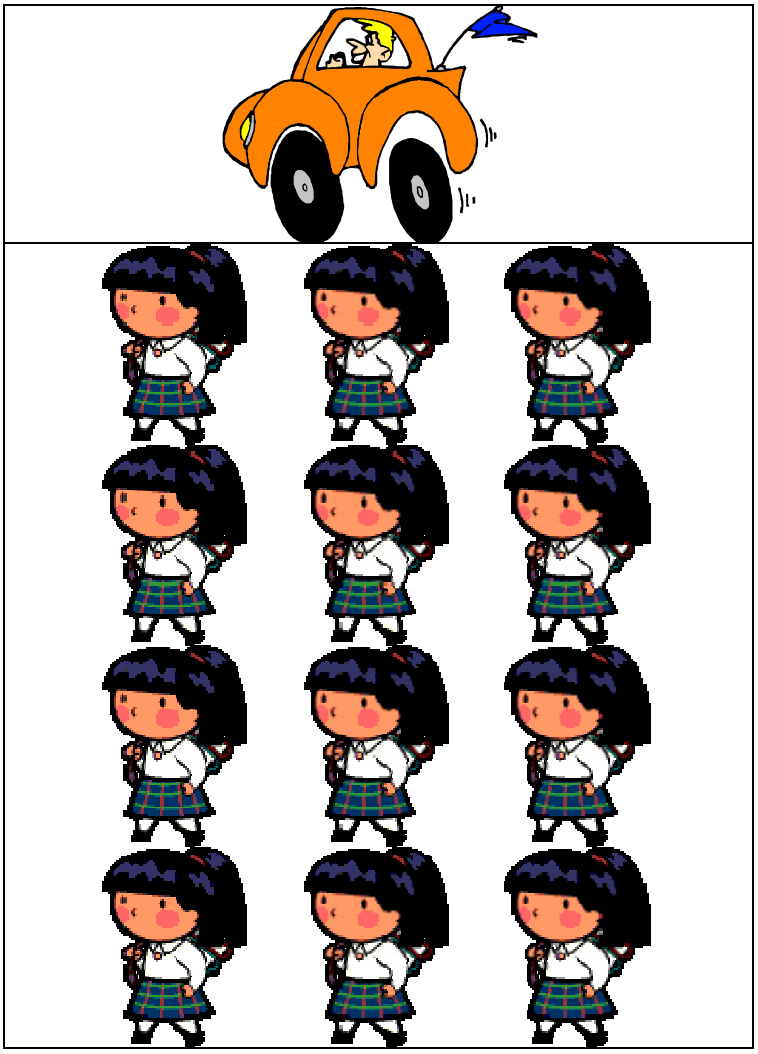
4) Draw a picture graph that shows the number of people that voted for their favourite animal

Animal	Number of people
Dog	10
Cat	8
Rabbit	2
Horse	4
Mouse	5
Chicken	4
Lion	5
Tiger	3
Snake	1
Monkey	0

5) Describe what this picture graph might be showing



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Year 3 Mathematics

Shapes

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Useful formulae and hints

2D (2 dimensional) shapes have a length and a width but no height. They can be thought of as “flat”.

The 2D shapes in this unit are

Triangles (3 sided)

Quadrilaterals (4 sided)

Pentagons (5 sided)

Quadrilaterals can be broken down into other groups:

- Parallelograms have two pairs of parallel sides. Each pair of sides are equal in length
- Rectangles are a special type of parallelogram whose sides form right angles with each other.
- Squares are a special type of rectangle where all 4 sides are equal in length
- Trapeziums have only one pair of parallel sides

There are also special groups of 3 and 5 sided shapes, but they are not looked at in this unit

Three dimensional (3D) shapes have height as well as length and width.

Some common 3D shapes are

- Cylinders

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- Pyramids
- Cones
- Rectangular prisms

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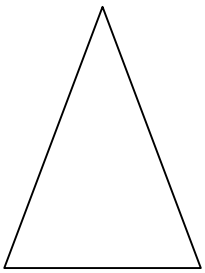
Exercise 1

Common 2D Shapes

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1) Name these shapes

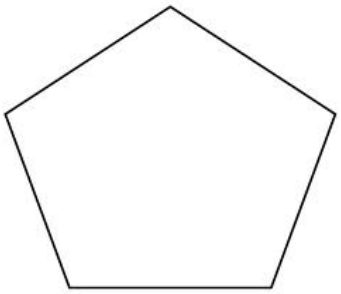
a)



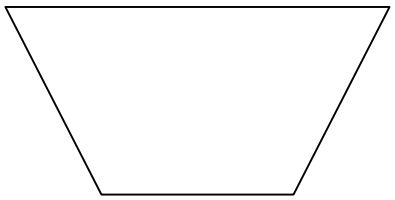
b)



c)



d)



e)

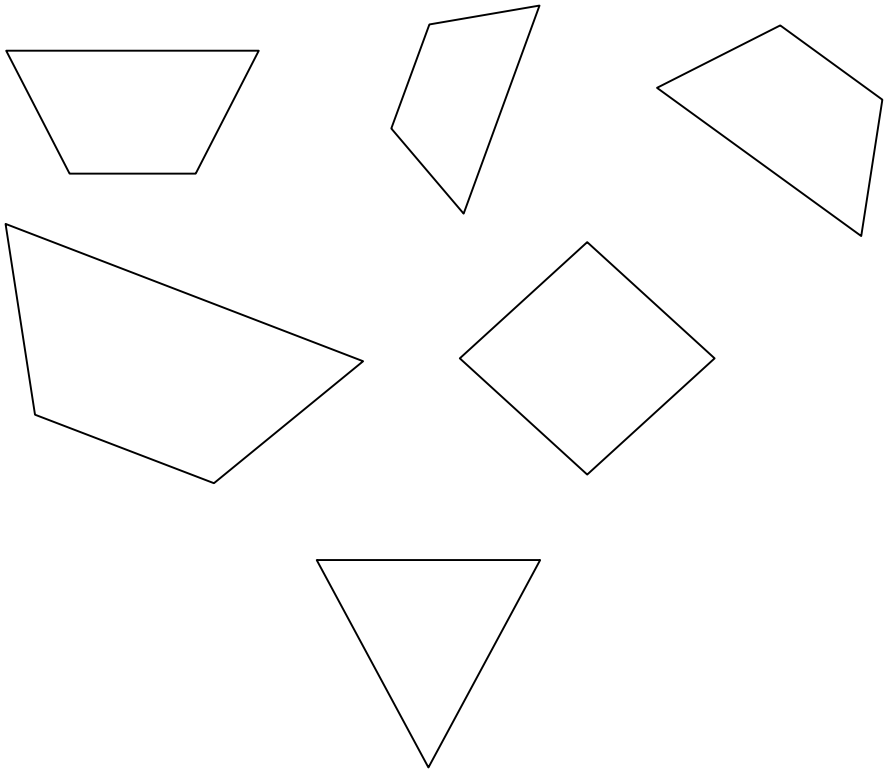


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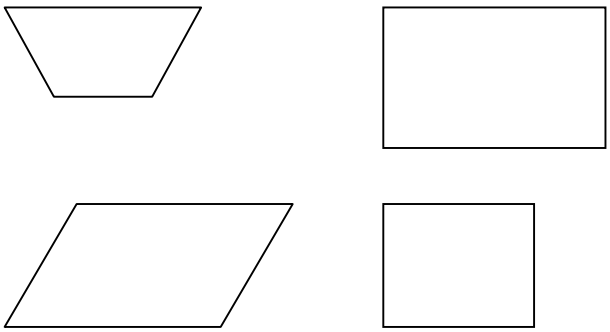
2) In the space in the table, write down how many sides each of the shapes has

Triangle	
Square	
Rectangle	
Parallelogram	
Pentagon	
Trapezium	

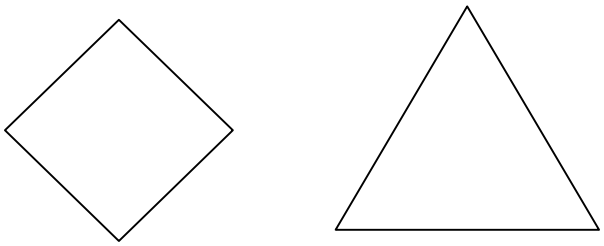
3) Which of the following shapes is NOT a trapezium?



4) Which of these shapes is NOT a parallelogram?



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5)

a) What do a square, rectangle, parallelogram and a trapezium all have in common?

b) What makes a trapezium different to the other three?

c) What makes a rectangle different to a parallelogram?

d) What makes a square different to a rectangle?

e) Do you think that a square is a special type of rectangle, or that a rectangle is a special type of square?

6)

Name the shape from the descriptions (could be more than one name)

a) It has 4 sides

b) It has 4 sides with both pairs of sides parallel

c) It has 4 sides with both pairs of sides parallel and all corners are right angles

d) It has 4 sides with both pairs of sides parallel and all corners are not right angles

e) It has 4 sides with both pairs of sides parallel, all sides the same length and all corners are right angles

f) It has 4 sides with one pair of sides parallel

g) It has 5 sides

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7) Name the shapes that you can see in this picture



How many squares are in this picture?

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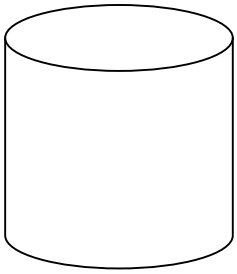
Exercise 2

Simple 3D Shapes

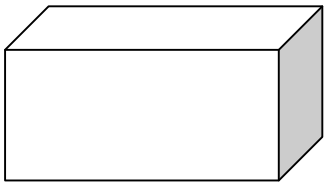
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1) Name these shapes

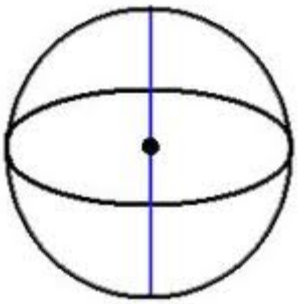
a)



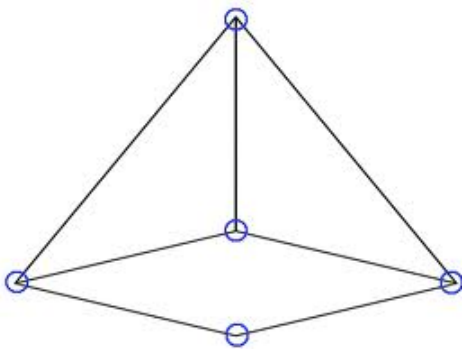
b)



c)

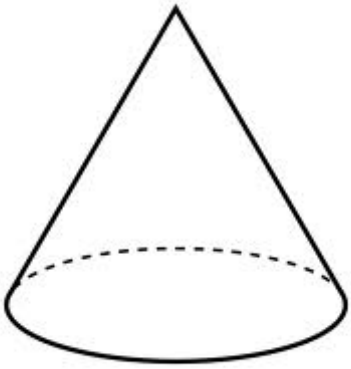


d)



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e)



2) What is the main difference between a cone and a pyramid?

3) Sort the following shapes into two groups; those that can have a square base and those that cannot

Pyramid, Sphere, Cone, Cylinder, Prism

4) Which of the above shapes can have a point at the top (an apex)?

5) Which of the above shapes best describes the following?

a) The shape of the Earth?

b) The shape of an ice cream?

c) The shape of a can of soup?

d) The shape of a cardboard box?

6) What is the same about a cone and a cylinder?

7) If you had a cylinder and a cone that had the same size base and were the same height which one could you fit more water into?

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8) Name the shape that these everyday objects are made from

a)



b)



c)



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d)



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Year 3 Mathematics

Measurement

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Useful formulae and hints

To read time on an analogue clock (one with hands)

The small hand indicates the hour (either just gone or approaching), while the big hand shows the number of minutes past or to the hour.

The numbers indicate the hour when the small hand is on them.

The numbers are also 5 minutes apart for the big hand. For example, if the big hand is on the 2, it is ten minutes past the hour

Fifteen minutes past the hour (big hand on the 3) is called quarter past, thirty minutes past the hour (big hand on the 6) is called half past, whilst fifteen minutes to the hour (big hand on the nine) is called quarter to. If the big hand is on the 12 it is o'clock

The kg is the unit of mass; there are 100g in a kg

The metre (m) is the unit of length; there are 100cm in a metre

The area of a shape is the number of square blocks of a certain size that covers the shape. The usual units of the blocks are 1 cm x 1 cm or 1 m x 1 m depending on the size of the shape

The volume of a shape is the number of cubes of a certain size that can fit exactly inside it. The usual size of the block is 1 cm x 1cm x 1cm

The amount of liquid that can fit inside a shape is called its capacity, and it is related to its volume. The unit of volume is the litre

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Exercise 1

Time

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1) What time is it on the following clocks?

a)



b)



c)



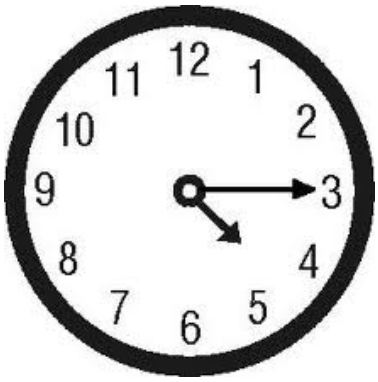
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d)



2) What time is it on the following clocks?

a)



b)

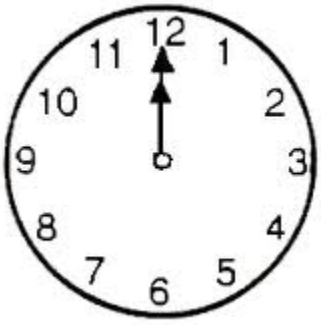


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c)

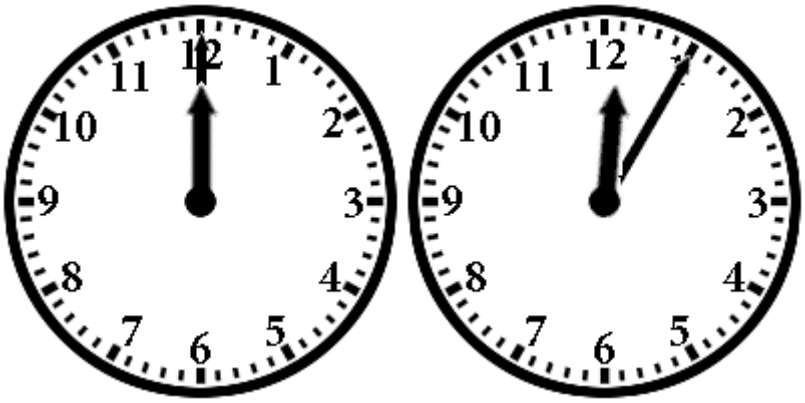


d)



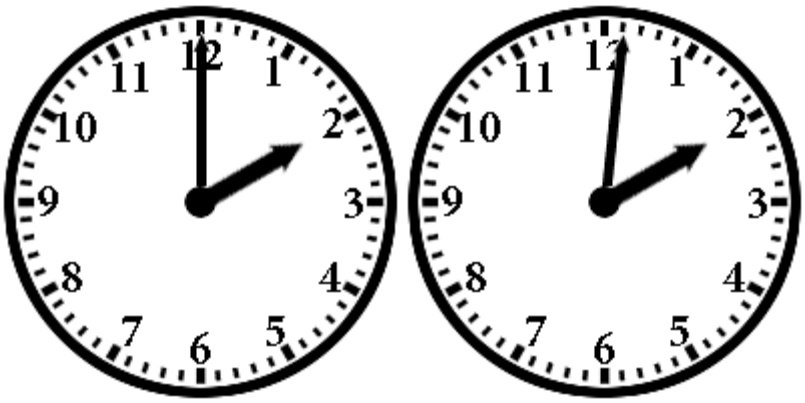
3) How much time has gone by between each of the two clocks?

a)

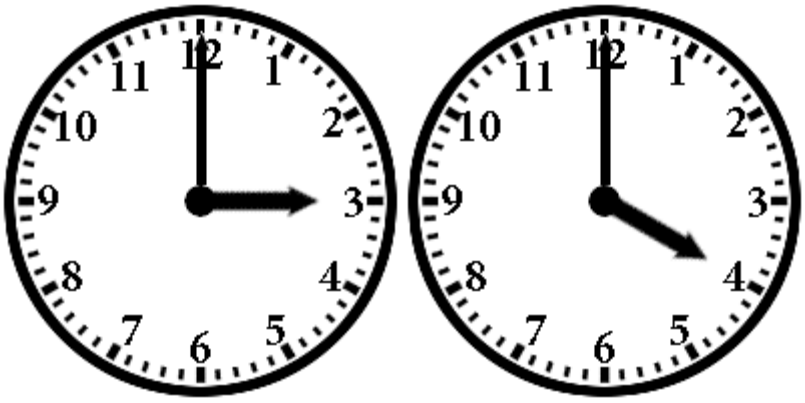


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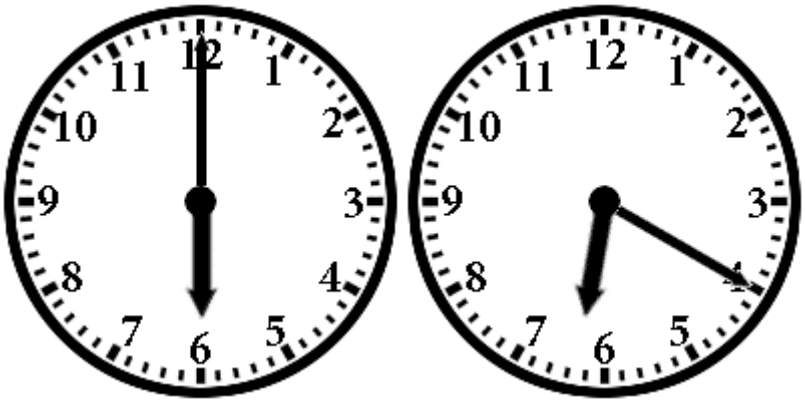
b)



c)



d)



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4) Draw the following times on a clock

a) One fifteen

b) Quarter to three

c) Half past two

d) Two thirty

e) Quarter past four

5) Peter started homework at 6 o'clock and had it all finished by half past 6. How much time had he spent on his homework?

6) Karen went to bed at eight fifteen, and Robert at seven forty five. How many minutes were there between their bedtimes?

7) John's dad leaves for work at seven o'clock in the morning and gets to work at half past seven. How long does it take John's dad to get to work?

8) Bill's mum was exercising. She started at five fifteen and had to exercise for 45 minutes. At what time should she stop exercising?

9) A worker gets fifteen minutes for morning tea. If he starts morning tea at 8:45, what time should he start work again?

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Exercise 2

Mass

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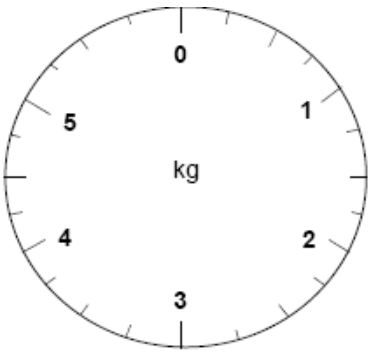
1) Which of the following would weigh about 1kg?

- One litre of water
- An elephant
- A car
- A baseball bat
- A fly
- Two tubs of margarine
- A person
- A biscuit
- Bag of sugar

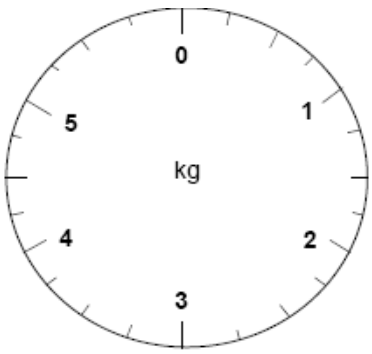
2) Tom has 3 objects that each weighs 1 kg. How much do the objects weigh in total?

3) Draw a pointer on the scale to show the following masses

a) 1 kg

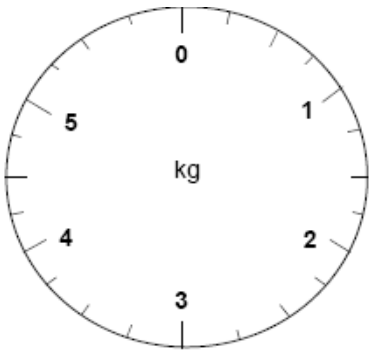


b) 2 kg

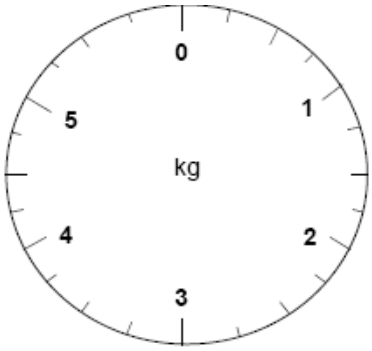


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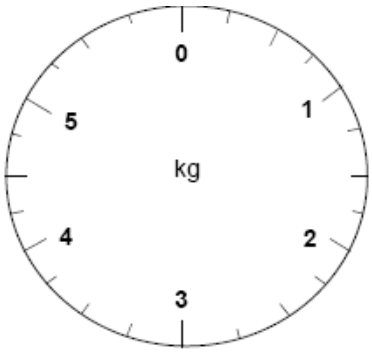
c) 4 kg



d) 3 and a half kg



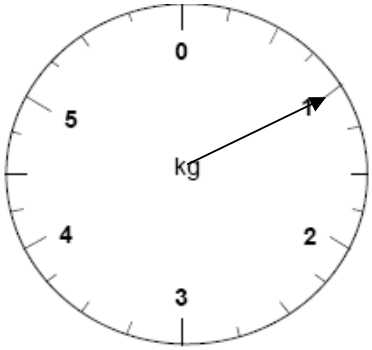
e) Half a kg



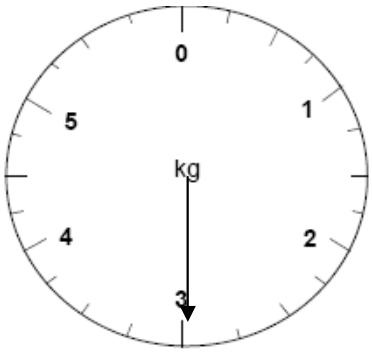
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4) Write down the reading on each scale

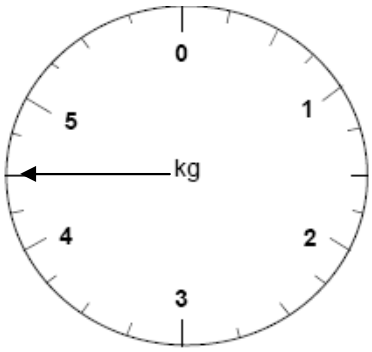
a)



b)

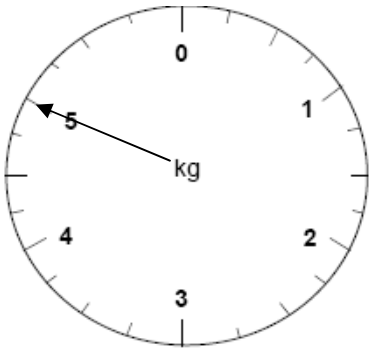


c)

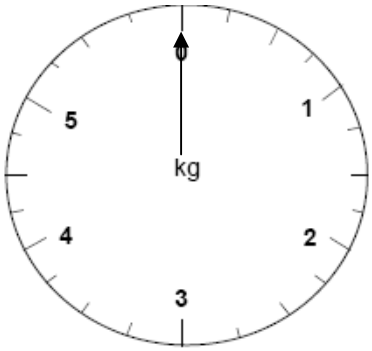


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d)



e)



5) Eric has some margarine tubs that each has a mass of ½ kg. How many would he need to place on a scale so it balances with objects that have a mass of:

a) 2 kg

b) 1 and ½ kg

c) 4 kg

d) 3 kg

6) What has more mass:

a) A truck or a bicycle?

b) A man or a giraffe?

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c) A fly or a plane?

d) A mobile phone or a computer?

e) A packet of biscuits or a carton of soft drink?

f) A piece of string or a jumper?

7) Mr Jones got two of his students to help move sand in a wheelbarrow. Mr Jones moved 25 kg, Robert moved 10 kg, and Alex moved 8 kg. How much sand did they move altogether?

8) Sausages cost $2 for 3 kg; how much would 6 kg of sausages cost?

9) Eric's father weighs 3 times as much as Eric. If his father weighs 75 kg, how much does Eric weigh?

10) John has one 1 kg weight, three 2 kg weights, and one 5 kg weight What would he need to place on a scale to balance:

a) 5 kg

b) 6 kg

c) 7 kg

d) 9 kg

e) 12 kg

f) 15 kg

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Exercise 3

Length

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1) How many centimetres in:

a) 1 meter

b) ½ metre

c) 2 metres

d) 3 ½ metres

e) 5 metres

2) How many metres in:

a) 50 cm

b) 100 cm

c) 200 cm

d) 250 cm

e) 550 cm

3) Graham is 145 cm tall, while his dad is 2 metres. How much taller is Graham's dad?

4) How many 1 meter rulers would need to be laid end to end to measure a length of 375 cm?

5) Mark rolled a ball 5m 40cm, while his friend rolled a ball 6m 10 cm. How much further did the second ball travel?

6) Geoffrey is painting a 4 metre line. A full paintbrush paints 20 cm before it has to be dipped back into the can. How many times does Geoffrey have to dip his brush in the can to paint the line?

7) Two pieces of wood are joined together length ways. The first piece is 2 m 60 cm long; the second piece is 3 m 60cm long. How long is the new piece of wood?

8) James took a pace and measured it to be 80 cm, while his dad's pace is 1m 30 cm; how far would they step one after the other?

9) Half the length of a piece of rope is 2m 75 cm. How long is the piece of rope?

10) A horse is 2m 70 cm tall, while its rider is 1m 50cm. How far off the ground is the rider when he is sitting on the horse?

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Exercise 4

Area

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1) The picture shows a block that is 1 cm long and 1 cm wide



a) What is the common name for this sized square?

b) Estimate how many of these squares it would take to fill

I. Your computer screen
II. Your exercise book
III. Your desk
IV. Your bedroom floor

2) How many cm^3 is shown in each of the following diagrams?

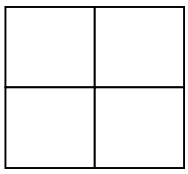
a)



b)

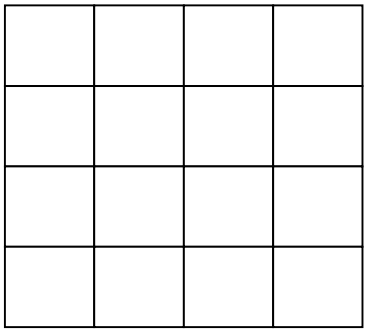


c)



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d)



e) Explain why the answers to parts a and c are the same although the total shape is different

3) Make, draw or imagine a grid with 10 rows and 10 columns of 1 cm squares. Which of the following would your grid be too small to fit, which would your grid be too big for, and which would it fit pretty close to exactly?

a) A school book

b) A floor

c) A back yard

d) The lid of a laptop

e) A calculator

f) A stamp

g) A chessboard

4) Put the following in order from smallest to largest area

- The roof of a car
- A plate
- An IPod

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- The side of a house
- A coin
- Australia

5) Peter made a grid of 1 cm squares. His grid was 5 rows by 4 columns. John made a grid that was 3 rows by 6 columns.

a) Whose grid was bigger and by how many squares?

b) Graham made a grid that was 10 rows and 2 columns. Whose grid was the same area as Graham's?

c) Graham's grid fitted exactly over a book; why wouldn't either of Peter's or John's grids fit exactly over the same book, even though one of them is the same area?

d) How many of Peter's grids would be needed to cover an object that was 20 rows long and 16 columns wide?

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Exercise 5

Volume

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1) Which of the following could hold 1 litre of liquid or more?

- A milk carton
- A car's petrol tank
- A teacup
- A bath
- A straw
- A swimming pool
- A teaspoon

2) Put the following in order from smallest to largest capacity

- A tablespoon
- A spa bath
- A dog's drinking bowl
- A shampoo bottle
- An ocean
- An eye dropper

3) Each of the small cubes measures 1cm x 1cm x 1cm.

a) What is the common name for such a cube?

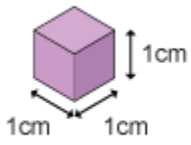
b) Why is a cube useful for measuring volumes of objects?

c) Name an object that would take about 12 cubes inside

d) Name an object that would be too big to measure the volume of by just using these cubes

4) Each cube is 1cm x 1cm x 1cm. What is the volume of each of the stacks?

a)



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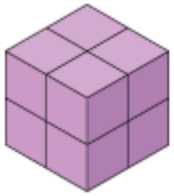
b)



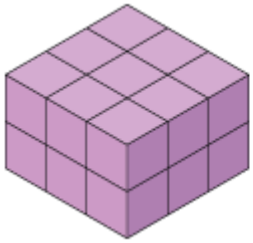
c) A



d)



e)



f) Explain why the answer to part c and d can be the same even though the stacks look different

5) Stacks of 1 cm blocks are built. How many blocks are in each of the following stacks?

a) 2 rows and 3 columns

b) 4 rows and 5 columns

c) 6 rows and 3 columns

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d) 3 rows and 6 columns

e) 10 rows and 10 columns

f) 30 rows and 30 columns

g) Which of the above stacks do you think would nearly fill a one litre container?

6) Mark had a stack of 1 cm blocks that was 5 rows and 6 columns. Peter's stack was twice as long and twice as wide. How many more blocks did Peter's stack have in it than Mark's?

7) Alan had a stack of 1 cm blocks that was 10 rows and 4 columns. David's stack had half as many rows and half as many columns. How many less blocks did David's stack have?

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Year 3 Mathematics

Space

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Useful formulae and hints

Maps are diagrams of places. They are drawn to scale so that the distance in real life can be worked out from the map.

Maps show such things as the location of streets, places of interest, buildings and parks.

By reading maps and following the scale and direction, a person can find places they need to go

Places on a map are shown by the use of a grid with letters or numbers across the top and down the side. Places can be found by matching the letters and numbers given to find the location

Example

	A	B	C	D
1				
2		School		
3	Football Ground			
4				Hospital

The football ground is at A4, the school is at B2, the hospital is at D4

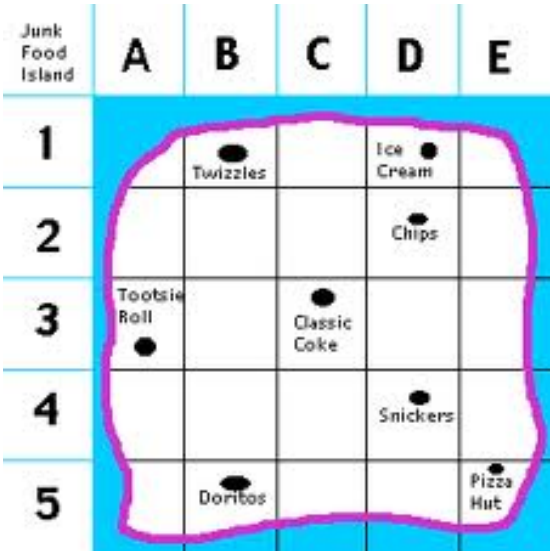
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Exercise 1

Map Legends & Directions

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1) Answer the following questions about Junk Food Island



a) Where would you find chips?

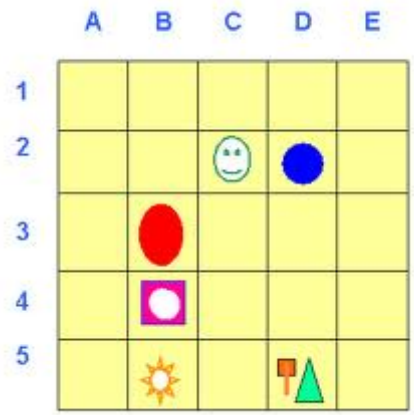
b) What is located at B1

c) Name a grid location where there is no junk food

d) Name a grid location where there is mostly water

e) If you started at Doritos and walked right until you came to the next grid location of junk food, what grid location would you be at?

2)



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a) At what grid location would you find a smiley face?

b) At what grid location would you find a triangle

c) What is at grid location B5?

d) Which row contains no symbols?

3) The map shows some streets in Sydney



a) Central Station is at the end of which street?

b) Which street joins Pitt St, George St and Sussex St?

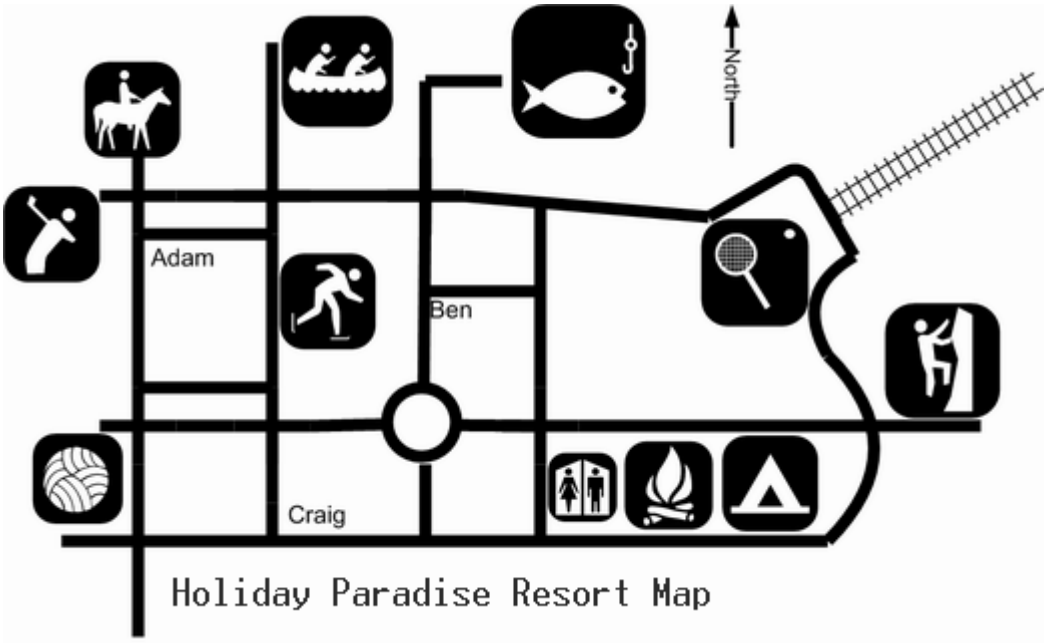
c) "We are here" on the corner of which two streets?

d) If you go left from "We are here" which street do you go down?

e) Which street is Chinatown near?

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4) Use the map of Holiday Paradise Resort to answer the questions



a) Whose house is closest to the golf?

b) If you started at Craig's house and walked north as far as you could go, what would you find?

c) What sport is played near the railway line?

d) Describe how you would get from horse riding to fishing (the quickest way)

e) Start at Ben's house facing north, walk north and turn left at the next street. Turn left at the next street and go all the way to the end. Where do you finish?

f) Even though it is not shown on the map, where do you think the lake is, and why?

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5) Record the following information on the grid

	A	B	C	D	E	F	G	H	I
1									
2									
3									
4									
5									
6									
7									
8									

a) There is a shop at A1

b) Three squares down from the shop is the police station

c) From the police station go toward the right of the page 4 squares and draw a school

d) A river runs from A8 to G8

e) There is a car park running from A7 to C7

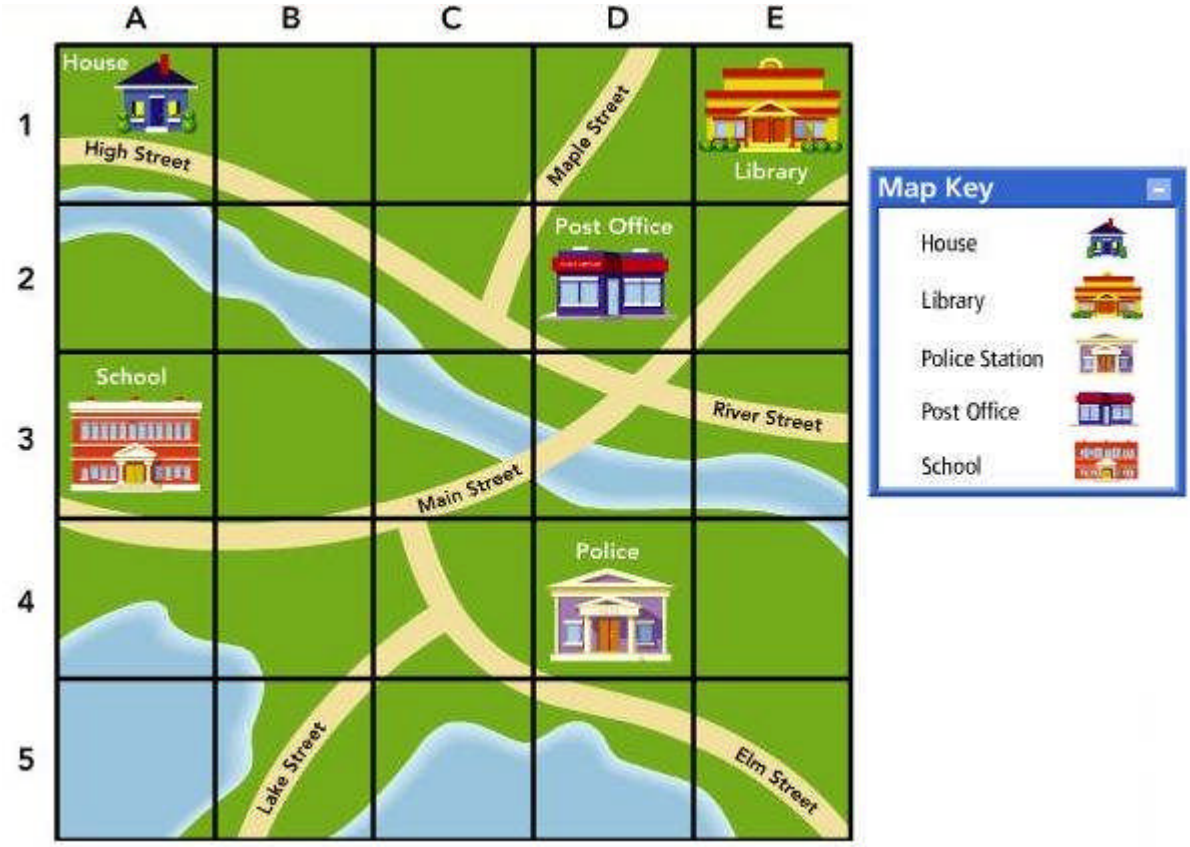
f) At H6 there is a restaurant

g) Draw roads that connect all the major parts of the map

h) On any part of the map separated from buildings or roads by one square or more draw parkland

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6)



a) Where is the post office located?

b) River Street changes its name to what street?

c) What building is at D4?

d) Name all the grid points that have at least part of the river in them

e) At what point does Lake Street meet Elm Street?

f) To get from the school to the library, what street would you need to take?

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